

Topic 19B: Nuclear magnetic resonance (NMR)
2. understand that ^{13}C NMR spectroscopy provides information about the positions of ^{13}C atoms in a molecule
3. be able to use data from ^{13}C NMR spectroscopy to: - i predict the different environments for carbon atoms present in a molecule, given values of chemical shift, δ - ii justify the number of peaks present in a ^{13}C NMR spectrum because of carbon atoms in different environments
4. understand that high resolution proton NMR provides information about the positions of ^1H atoms in a molecule
5. be able to use data from high resolution ^1H NMR spectroscopy to: - i predict the different types of proton present in a molecule, given values of chemical shift, δ - ii relate relative peak areas, or ratio numbers of protons, to the relative numbers of ^1H atoms in different environments - iii deduce the splitting patterns of adjacent, non-equivalent protons using the (n+1) rule and hence suggest the possible structures for a molecule - iv predict the chemical shifts and splitting patterns of the ^1H atoms in a given molecule